

DentalOS — The AI Operating System for Dental Care Intelligence

“The \$550B dental industry runs on paper, intuition, and chaos. We’re building the intelligence layer that transforms every practice into a precision medicine powerhouse.”

The Billion-Dollar Problem

The global dental industry is a **\$550 billion market** stuck in the 1990s:

- **90% of dental diagnoses** rely on visual inspection and subjective interpretation
- **43% of cavities** are missed by human examination alone
- **\$50B+ annually** lost to inefficient treatment planning and insurance denials
- **Average dental practice** uses 7+ disconnected software systems
- **70% of adults** avoid the dentist due to cost uncertainty and fear

Meanwhile: - AI can detect cavities with **95%+ accuracy** from X-rays - Predictive models can forecast decay, gum disease, and oral cancer years in advance - Insurance denials can be reduced by **60%** with proper documentation - Patient engagement AI can increase treatment acceptance by **40%**

The dental industry is the last major healthcare vertical without an AI operating system.

The Vision: DentalOS

DentalOS is the **unified AI intelligence platform** that transforms dental practices from reactive treatment centers into **predictive oral health engines**.

Core Platform Components

1. DiagnosticAI Engine

- Real-time analysis of X-rays, CBCT scans, intraoral photos
- Detection: cavities, periodontal disease, bone loss, oral cancer, TMJ disorders
- Confidence scoring and attention mapping
- Longitudinal tracking across patient history

2. TreatmentMind

- AI-generated treatment plans optimized for outcomes + patient preferences
- Multi-scenario modeling (conservative vs. aggressive approaches)
- Cost estimation with insurance coverage prediction
- Phased treatment sequencing

3. InsuranceAI

- Automatic pre-authorization with optimal documentation
- Claim denial prediction and prevention
- Appeal automation with 85%+ success rate
- Fee schedule optimization across payers

4. PatientBridge

- Conversational AI for scheduling, reminders, and education
- Treatment acceptance optimization through personalized communication
- Fear/anxiety detection and intervention protocols
- Financial planning and payment optimization

5. PracticeOS

- Unified dashboard across all practice systems
- Staff scheduling and productivity optimization
- Supply chain prediction and auto-ordering
- Revenue cycle management and analytics

6. OralGenome

- Integration with genetic risk factors for caries, periodontal disease
 - Microbiome analysis for personalized prevention
 - Saliva biomarker monitoring
 - Whole-body health correlation (oral-systemic link)
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Market Opportunity

Total Addressable Market (TAM): \$45B

- Global dental practice management software: \$3.5B
- Dental imaging/diagnostics: \$4B
- Dental insurance/RCM software: \$2B
- **AI-native replacement opportunity: \$45B**

Serviceable Addressable Market (SAM): \$18B

- North America + Europe practices: 350,000
- Average potential spend per practice: \$50K/year
- **SAM: \$17.5B annually**

Serviceable Obtainable Market (SOM): \$1.8B (Year 5)

- Target: 36,000 practices (10% penetration)
 - Average revenue per practice: \$50K/year
 - **SOM: \$1.8B ARR**
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Business Model

Pricing Tiers

Tier	Price	Features
Starter	\$499/mo	DiagnosticAI + PatientBridge
Professional	\$999/mo	+ TreatmentMind + InsuranceAI
Enterprise	\$2,499/mo	Full platform + OralGenome + API
DSO/Group	Custom	Multi-location + analytics + white-label

Revenue Streams

1. **SaaS Subscriptions** (70% of revenue)
 - Per-practice monthly licensing
 - Tiered by features and volume
2. **Transaction Fees** (20% of revenue)
 - Insurance claim processing: \$2-5/claim

- Patient financing facilitation: 1% of financed amount
 - Lab order routing: \$3-5/case
3. **Data & Insights** (10% of revenue)
- Anonymized benchmarking data for DSOs
 - Research partnerships with dental schools
 - Pharma/device company market intelligence

Competitive Landscape

Current Players (Legacy)

Company	Weakness
Dentrix/Henry Schein	30-year-old architecture, no AI
Eaglesoft/Patterson	Fragmented, poor UX, no intelligence
Open Dental	Open-source chaos, no unified AI
Curve Dental	Cloud but shallow AI capabilities
Pearl	Point solution (diagnostics only)

DentalOS Differentiation

- **Full-stack platform** vs. point solutions
 - **AI-native architecture** vs. bolt-on features
 - **Unified data model** vs. integration hell
 - **Outcome-based insights** vs. transaction records
 - **Consumer-grade UX** for dental professionals
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Go-to-Market Strategy

Phase 1: Beachhead (Months 1-12)

Target: Solo practices and small groups (1-5 locations)

- Partner with 5 dental schools for validation studies
- Launch DiagnosticAI with free tier (X-ray analysis)
- Build case studies showing ROI (missed diagnoses, insurance acceptance)
- Focus: California, Texas, Florida (highest practice density)

Phase 2: Expansion (Months 12-24)

Target: Mid-market DSOs (6-50 locations)

- Launch full platform with TreatmentMind + InsuranceAI
- Build DSO-specific analytics dashboard
- Partner with Henry Schein/Patterson for distribution
- Establish dental conferences presence (ADA, Greater NY)

Phase 3: Enterprise (Months 24-36)

Target: Large DSOs (50+ locations) and dental networks

- White-label offering for major DSOs
- Integration partnerships with major imaging hardware
- Launch OralGenome with lab partnerships

- International expansion: UK, Germany, Australia

Phase 4: Platform (Months 36-48)

Target: Ecosystem dominance

- API marketplace for dental tech
- Patient-facing consumer app (oral health score)
- Dental insurance carrier partnerships
- Acquisition of complementary point solutions

Technology Architecture

AI Model Stack

DentalOS Intelligence Layer

Vision AI	NLP Engine	Predictive Models
- X-ray	- Notes	- Decay forecast
- CBCT	- Claims	- Perio risk
- Photos	- Patient	- Treatment outcome

Unified Dental Data Model

(Patient → Diagnosis → Treatment → Outcome → Learning)

Integration Layer (FHIR Dental)

Imaging PMS Insurance Labs Hardware EHR

Data Moat

- **Training data:** Partner with 5 academic medical centers for 10M+ annotated dental images
- **Outcome data:** Track treatment outcomes across network for continuous learning
- **Longitudinal data:** Build patient histories across practice changes
- **Network effects:** More practices → better benchmarks → more value → more practices

Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Practices	500	3,000	12,000	28,000	50,000
ARR	\$3M	\$24M	\$120M	\$350M	\$750M
Gross Margin	65%	72%	78%	80%	82%
CAC	\$5,000	\$4,000	\$3,500	\$3,000	\$2,500
LTV	\$45K	\$55K	\$65K	\$75K	\$85K
LTV:CAC	9x	14x	19x	25x	34x

Team Requirements

Founding Team (Raise: Seed)

- **CEO:** Healthcare SaaS veteran, ideally dental industry experience
- **CTO:** ML/AI platform builder, medical imaging background
- **Chief Dental Officer:** Practicing dentist + dental school faculty
- **VP Engineering:** Health tech infrastructure, HIPAA expertise

Key Hires (Series A)

- VP Sales: Dental/healthcare vertical sales leader
- VP Product: Consumer health UX expertise
- VP Clinical: DSO operations background
- Head of AI: Computer vision specialist

Risk Factors & Mitigations

Risk	Mitigation
Regulatory (FDA)	Start with decision support (not diagnosis), pursue 510(k) for diagnostic claims
Incumbent response	Move fast, build data moat, focus on AI-native vs. bolt-on
Practice adoption	Freemium model, prove ROI before full commitment
Data privacy	HIPAA-first architecture, patient consent framework
Integration complexity	Open API, partner with legacy vendors

Investment Ask

Seed Round: \$5M

Use of Funds: - Product development (60%) - Clinical validation studies (20%) - Initial GTM team (15%)
- Regulatory/legal (5%)

Series A: \$25M (Month 18)

Milestones: - 500+ practices on platform - \$3M ARR - DiagnosticAI FDA pathway defined - 2+ DSO pilot programs

Series B: \$80M (Month 36)

Milestones: - 12,000+ practices - \$120M ARR - International expansion - OralGenome launch

Exit Scenarios

Scenario	Valuation	Likely Acquirers
Strategic Acquisition (Year 4-5)	\$2-4B	Henry Schein, Dentsply Sirona, Envista
PE Rollup (Year 5-6)	\$3-5B	KKR, Blackstone, Warburg Pincus
IPO (Year 6-7)	\$5-10B	Public markets

Comparable transactions: - Heartland Dental acquired by KKR: \$2.8B - Pearl AI raised \$75M at \$500M+ valuation (diagnostics only) - Dental Intelligence acquired by Henry Schein: undisclosed

The Ask

We're building the **Salesforce for Dentistry** — the unified AI platform that every dental practice needs but doesn't exist yet.

The \$550B dental industry is: - **Fragmented:** 200,000+ practices in the US alone - **Inefficient:** Running on 30-year-old software - **Ripe for AI:** Massive imaging data, clear diagnostic criteria

DentalOS will capture this market by being first to deliver: - True AI-native diagnostics - Unified practice intelligence - Measurable ROI from day one

The mouth is the gateway to the body. DentalOS is the gateway to AI-transformed oral healthcare.

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For investment inquiries and partnership opportunities